weights in a chemical similarity matrix data-mined from the ICSD [45].

The WBM authors performed 5 iterations of this substitution process (we refer to these steps as batches). After each step, the newly generated structures found to be thermodynamically stable after DFT relaxation flow back into the source pool to partake in the next round of substitution. This split of the data into batches of increasing substitution count is a unique and compelling feature of the test set as it allows out-of-distribution (OOD) testing by examining whether model performance degrades for later batches. A higher number of elemental substitutions on average carries the structure further away from the region of material space covered by the MP training set (see fig. S6 for details). Whilst this batch information makes the WBM dataset an exceptionally useful data source for examining the extrapolation performance of ML models, we look primarily at metrics that consider all batches as a single test set.

In order to control for the potential adverse effects of leakage between the MP training set and the WBM test set, we cleaned the WBM test set based on protostructure matching. We refer to the combination of a materials prototype and the elemental assignment of its wyckoff positions as a protostructure following [46]. First we removed 524 pathological structures in WBM based on formation energies being larger than 5 eV/atom or smaller than -5 eV/atom. We then removed from the WBM test set all examples where the final protostructure of a WBM material matched the final protostructure of an MP material. In total 11,175 materials were cleaned using this filter. We further removed all duplicated protostructures within WBM, keeping the lowest energy structure in each instance, leaving 215,488 structures in the unique prototype test set.

Throughout this work, we define stability as being on or below the convex hull of the MP training set ($E_{\text{MP hull dist}} \leq 0$). 32,942 out of 215,488 materials in WBM unique prototype test set satisfy this criterion. Of these, ~33k are unique prototypes, meaning they have no matching structure prototype in MP nor another higher-energy duplicate prototype in WBM. Our code treats the stability threshold as a dynamic parameter, allowing for future model comparisons at different thresholds. For initial analysis in this direction, see fig. S1 in the SI.

As WBM explores regions of materials space not well sampled by MP, many of the discovered materials that lie below MP's convex hull are not stable relative to each other. Of the ~33k that lie below the MP convex hull less than half, or around ~20k, remain on the joint MP+WBM convex hull. This observation suggests that many WBM structures are repeated samples in the same new chemical spaces. It also highlights a critical aspect of this benchmark in that we knowingly operate on an incomplete convex hull. Only current knowledge of competing points on the PES is accessible to a real discovery campaign and our metrics are designed to reflect this.

2.1.3 Limitations of this Framework

Whilst the framework proposed here directly mimics a common computational materials discovery workflow, it is worth highlighting that there still exist significant limitations to these traditional computational workflows that can prevent the material candidates suggested by such a workflow from being able to be synthesized in practice. For example, high-throughput DFT calculations often use small unit cells which can lead to artificial orderings of atoms. The corresponding real material may be disordered due to entropic effects that cannot be captured in the zero-kelvin thermodynamic convex hull approximated by DFT [47].

Another issue is that, when considering small unit cells, the DFT relaxations may get trapped at dynamically unstable saddle points in the true PES. This failure can be detected by calculating the phonon spectra for materials predicted to be stable. However, the cost of doing so with DFT is often deemed prohibitive for high-throughput screening. The lack of information about the dynamic stability of nominally stable materials in the WBM test set prevents this work from considering this important criteria as an additional screening filter. However, recent progress in the development of UIPs suggests that ML approaches will soon provide sufficiently cheap approximations of these terms for high-throughput searches [48, 49]. As the task presented here begins to saturate, we believe that future discovery benchmarks should extend upon the framework proposed here to also incorporate criteria based on dynamic stability.

When training UIP models there is a competition between how well given models can fit the energies, forces, and stresses simultaneously. The metrics in the Matbench Discovery leaderboard are skewed towards energies and consequently UIP models trained with higher weighting on energies can achieve better metrics. We caution that optimizing hyperparameters purely to improve performance on this benchmark may have unintended consequences for models intended for general purpose use. Practitioners should also consider other involved evaluation frameworks that explore orthogonal use cases when developing model architectures. We highlight work from Póta *et al.* [50] on thermal conductivity benchmarking, Fu *et al.* [51] on MD stability for molecular simulation, and Chiang [52] on modeling reactivity (hydrogen combustion) and asymptotic correctness (homonuclear diatomic energy curves) as complementary evaluation tasks for assessing the performance of UIP models.

We design the benchmark considering a positive label for classification as being on or below the convex hull of the MP training set. An alternative formulation would be to say that materials in WBM that are below the MP convex hull but do not sit on the combined MP+WBM convex hull are negatives. The issue with such a design is that it involves unstable evaluation metrics. If we consider the performance against the final combined convex hull rather than the initial MP convex hull, then each additional sample considered can retroactively change whether or not a previous candidate would be labeled as a success as it may no-longer sit on the hull. Since constructing the convex hull is computationally expensive, this path dependence makes it impractical to evaluate cumulative precision metrics (see Figure 2). The chosen setup does increase the number of positive labels and could consequently be interpreted as overestimating the performance. This overestimation decreases as the convex hull becomes better sampled. Future benchmarks building on this work could make use of combination of MP+WBM to control this artifact. An alternative framework could report metrics for each WBM batch in turn and retrain between batches, this approach was undesirable here as it increases the cost of submission five-fold and introduces many complexities, for example, should each model only retrain on candidates it believed to be positive, that would make fair comparison harder.

2.2 Models

To test a wide variety of methodologies proposed for learning the potential energy landscape, our initial benchmark release includes 13 models. Next to each model's name we give the training targets that were used: E - Energy, F - Forces, S - Stresses and M - Magnetic moments. The subscripts G and D refer to whether gradient-based or direct prediction methods were used to obtain force and stress predictions.

1. **EquiformerV2 + DeNS** [53, 54] (EFS_D) - EquiformerV2 builds on the first Equiformer model [55] by replacing the $SO(3)$ convolutions with eSCN convolutions [56] as well as a range of additional tweaks to make better use of the ability to scale to higher $L_{\max}$ using eSCN convolutions. EquiformerV2 uses direct force prediction rather than taking the forces as the derivative of the energy predictions for computational efficiency. Here we take the pre-trained "eqV2 S DeNS" [57] trained on the MPTrj dataset. This model in addition to supervised training on energies, forces, and stresses makes use of a auxilary training task based on de-noising non-equilibrium structures [54]. We refer to this model as "EquiformerV2 + DeNS" in the text and "eqV2 S DeNS" in plots.

2. **Orb** [58] (EFS_D) - Orb is a lightweight model architecture developed to scale well for the simulation of large systems such as metal organic frameworks. Rather than constructing an architecture that is equivariant by default, Orb instead makes use of data augmentation during training to achieve approximate equivariance. This simplifies the architecture allowing for faster inference. We report results for the "orb-mptrj-only-v2" model which was pre-trained using a diffusion-based task on MPTrj before supervised training on the energies, forces and stresses in MPTrj. For simplicity we refer to this model as "ORB MPTrj".

3. **SevenNet** [59] (EFS_G) - SevenNet emerged from an effort to improve the performance of message passing neural networks [60] when used to conduct large scale simulations involving that benefit from parallelism via spatial decomposition. Here we use the pre-trained "SevenNet-0_11July2024" trained on the MPTrj dataset. The SevenNet-0 model is an equivariant architecture based on a NequIP [61] architecture that mostly adopts the GNoME [62] hyperparameters. SevenNet-0 differs from NequIP and GNoME by replacing the tensor product in the self-connection layer with a linear layer applied directly to the node features, this reduces the number of parameters from 16.24 million in GNoME to 0.84 million for SevenNet-0. For simplicity we refer to this model as "SevenNet".

4. **MACE** [24] (EFS_G) - MACE builds upon the recent advances [61, 63] in equivariant neural network architectures by proposing an approach to computing high-body-order features efficiently via Atomic Cluster Expansion [64]. Unlike the other UIP models considered MACE was primarily developed for molecular dynamics of single material systems and not the universal use case studied here. The authors trained MACE on the MPTrj dataset, these models have been shared under the name "MACE-MP-0" [48] and we report results for the "2023-12-03" version commonly called "MACE-MP-0 (medium)". For simplicity we refer to this model as "MACE".

5. **CHGNet** [23] ($\mathrm{EFS}_G\mathrm{M}$) - CHGNet is a UIP for charge-informed atomistic modeling. Its distinguishing feature is that it was trained to predict magnetic moments on top of energies, forces and stresses in the MPtrj dataset (which was prepared for the purposes of training CHGNet). By modeling magnetic moments, CHGNet learns to accurately represent the orbital occupancy of electrons which allows it to predict both atomic and electronic degrees of freedom. We make use of the pre-trained "v0.3.0" CHGNet model from [23].

6. **M3GNet** [22] (EFS_G) - M3GNet is a GNN-based UIP for materials trained on up to 3-body interactions in the initial, middle and final frame of MP DFT relaxations. The model takes the unrelaxed input and emulates structure relaxation before predicting energy for the pseudo-relaxed structure. We make use of the pre-trained "v2022.9.20" M3GNet model from [22] trained on the compliant MPF.2021.2.8 dataset.

7. **ALIGNN** [65] (E) - The Atomistic Line Graph Neural Network (ALIGNN) is a message passing GNN

Model	F1 ↑	DAF ↑	Prec ↑	Acc ↑	TPR ↑	TNR ↑	MAE ↓	RMSE ↓	R^2 ↑	Training Set	Model Params	Targets	Date Added
eqV2 S DeNS	0.815	5.042	0.771	0.941	0.864	0.953	0.036	0.085	0.788	146K (1.6M) (MPtrj)	31.2M	EFS_D	2024-10-18
ORB MPtrj	0.765	4.702	0.719	0.922	0.817	0.941	0.045	0.091	0.756	146K (1.6M) (MPtrj)	25.2M	EFS_D	2024-10-14
SevenNet	0.724	4.252	0.650	0.904	0.818	0.919	0.048	0.092	0.750	146K (1.6M) (MPtrj)	842.4K	EFS_G	2024-07-13
MACE	0.669	3.777	0.577	0.878	0.796	0.893	0.057	0.101	0.697	146K (1.6M) (MPtrj)	4.7M	EFS_G	2023-07-14
CHGNet	0.613	3.361	0.514	0.851	0.758	0.868	0.063	0.103	0.689	146K (1.6M) (MPtrj)	412.5K	EFS_GM	2023-03-03
M3GNet	0.569	2.882	0.441	0.813	0.803	0.813	0.075	0.118	0.585	63K (188.3K) (MPF)	227.5K	EFS_G	2022-09-20
ALIGNN	0.567	3.206	0.490	0.841	0.672	0.872	0.093	0.154	0.297	155K (MP 2022)	4.0M	Energy	2023-06-02
MEGNet	0.510	2.959	0.452	0.826	0.585	0.870	0.130	0.206	-0.248	133K (MP Graphs)	167.8K	Energy	2022-11-14
CGCNN	0.507	2.855	0.436	0.818	0.605	0.857	0.138	0.233	-0.603	155K (MP 2022)	128.4K (N=10)	Energy	2022-12-28
CGCNN+P	0.500	2.563	0.392	0.786	0.693	0.803	0.113	0.182	0.019	155K (MP 2022)	128.4K (N=10)	Energy	2023-02-03
Wrenformer	0.466	2.256	0.345	0.745	0.719	0.750	0.110	0.186	-0.018	155K (MP 2022)	5.2M (N=10)	Energy	2022-11-26
BOWSR	0.423	1.964	0.300	0.712	0.718	0.693	0.118	0.167	0.151	133K (MP Graphs)	167.8K	Energy	2022-11-17
Voronoi RF	0.333	1.579	0.241	0.668	0.535	0.692	0.148	0.212	-0.329	155K (MP 2022)	26.2M	Energy	2022-11-26
Dummy	0.185	1.000	0.154	0.687	0.232	0.769	0.124	0.184	0.000				

Table 1: Classification and regression metrics for all models tested on our benchmark ranked by F1 score. The heat map ranges from yellow (best) to blue (worst) performance. DAF = discovery acceleration factor is the ratio of model precision to percentage of stable structures in the test set, TPR = true positive rate, TNR = true negative rate, MAE = mean absolute error, RMSE = root mean squared error. The dummy classifier uses the `scikit-learn stratified` strategy of randomly assigning stable or unstable labels according to the training set prevalence. The dummy regression metrics MAE, RMSE and R^2 are attained by always predicting the test set mean. The top positions in the leaderboard are all taken by universal interatomic potential models trained on the combination of energies, forces and stresses. There is a pronounced gap in the regression metrics between the UIP models and the 7 energy-only models. It is worth noting that CGCNN+P, Wrenformer, and BOWSR achieve lower regression metrics through their mitigation strategies for initial and relaxed-structure mismatch but ultimately these strategies did not improve their usefulness as measured by the F1 score and DAF. Voronoi RF, CGCNN and MEGNet perform worse than dummy in regression metrics but better than dummy on some classification metrics, demonstrating that regression metrics alone can be misleading.

architecture that takes as input both the interatomic bond graph and a line graph corresponding to 3-body bond angles. The ALIGNN architecture involves a global pooling operation which means that it is ill-suited to force-field applications. To address this the ALIGNN-FF model was later introduced without global pooling [66]. We trained ALIGNN on the MP-crystals-2022.10.28 dataset for this benchmark.

8. **MEGNet** [44] (E) - MatErials Graph Network is another GNN-based architecture that also updates a set of edge and global features (like pressure and temperature) in its message passing operation. This work showed that learned element embeddings encode periodic chemical trends and can be transfer-learned from large datasets (formation energies) to predictions on small data properties (band gaps, elastic moduli). We make use of the pre-trained "Eform_MP_2019" MEGNet model trained on the compliant MP-crystals-2019.4.1 dataset.

9. **CGCNN** [67] (E) - The Crystal Graph Convolutional Neural Network (CGCNN) was the first neural network model to directly learn 8 different DFT-computed material properties from a graph representing the atoms and bonds in a periodic crystal. CGCNN was among the first to show that just like in other areas of ML, given large enough training sets, neural networks can learn embeddings that outperform human-engineered structure features directly from the data. We trained an ensemble of 10 CGCNN models on the MP-crystals-2022.10.28 dataset for this benchmark.

10. **CGCNN+P** [68] (E) - This work proposes simple, physically motivated structure perturbations to augment stock CGCNN's training data of relaxed structures with structures resembling unrelaxed ones but mapped to the same DFT final energy. Here we chose $P = 5$, meaning the training set is augmented with 5 random perturbations of each relaxed MP structure mapped to the same target energy. In contrast to all other structure-based GNNs considered in this benchmark, CGCNN+P is not attempting to learn the Born-Oppenheimer potential energy surface. The model is instead taught the PES as a step-function that maps each valley to its local minimum. The idea is that during testing on unrelaxed structures, the model will predict the energy of the nearest basin in the PES. The authors confirm this by demonstrating a lowering of the energy error on unrelaxed structures. We trained an ensemble of 10 CGCNN+P models on the MP-crystals-2022.10.28 dataset for this bench-

mark.

11. **Wrenformer** (E) - For this benchmark, we introduce Wrenformer which is a variation on the coordinate-free Wren model [11] constructed using standard QKV-self-attention blocks [69] in place of message-passing layers. This architectural adaptation reduces the memory usage allowing the architecture to scale to structures with greater than 16 Wyckoff positions. Like its predecessor, Wrenformer is a fast coordinate-free model aimed at accelerating screening campaigns where even the unrelaxed structure is a priori unknown [46]. The key idea is that by training on the coordinate anonymized Wyckoff positions (symmetry-related positions in the crystal structure), the model learns to distinguish polymorphs while maintaining discrete and computationally enumerable inputs. The central methodological benefit of an enumerable input is that it allows users to predict the energy of all possible combinations of spacegroup and Wyckoff positions for a given composition and maximum unit cell size. The lowest-ranked protostructures can then be fed into downstream analysis or modeling. We trained an ensemble of 10 Wrenformer models on the MP-crystals-2022.10.28 dataset for this benchmark.

12. **BOWSR** [21] (E) - BOWSR combines a symmetry-constrained Bayesian optimizer (BO) with a surrogate energy model to perform an iterative exploration-exploitation-based search of the potential energy landscape. Here we use the pre-trained "Eform_MP_2019" MEGNet model [44] for the energy model as proposed in the original work. The high sample count needed to explore the PES with BO makes this by far the most expensive model tested.

13. **Voronoi RF** [70] (E) - A random forest trained to map a combination of composition-based Magpie features [71] and structure-based relaxation-robust Voronoi tessellation features (effective coordination numbers, structural heterogeneity, local environment properties, ...) to DFT formation energies. This fingerprint-based model predates most deep learning for materials but significantly improved over earlier fingerprint-based methods such as the Coulomb matrix [72] and partial radial distribution function features [73]. It serves as a baseline model to see how much value the learned featurization of deep learning models can extract from the increasingly large corpus of available training data. We trained Voronoi RF on the MP-crystals-2022.10.28 dataset for this benchmark.

3 Results

Table 1 shows performance metrics for all models included in the initial release of Matbench Discovery reported on the unique protostructure subset. EquiformerV2 + DeNS takes the top spot and emerges as the current SOTA for ML-guided materials discovery. It outperforms the other models on all 9 reported metrics. When computing metrics in the presence of missing values or obviously pathological predictions (error of 5 eV/atom or greater) we assign the dummy regression values and a negative classification prediction to these points. The discovery acceleration factor (DAF) measures how many more stable structures a model found compared to the dummy discovery rate achieved by randomly selecting test set crystals. Formally the DAF is the ratio of the precision to the prevalence. The maximum possible DAF is the inverse of the prevalence which on our dataset is $(33\text{k}/215\text{k})^{-1} \approx 6.5$. Thus the current SOTA of 5.04 achieved by EquiformerV2 + DeNS leaves room for improvement. However, evaluating each model on the subset of 10k materials each model ranks as being most stable (see table S2), we see an impressive DAF of 6.33 for EquiformerV2 + DeNS which is approaching optimal performance for this task.

We find a large performance gap between models that make energy-only predictions directly from unrelaxed inputs (MEGNet, Wrenformer, CGCNN, CGCNN+P, ALIGNN, Voronoi RF) compared to UIPs that learn from force and stress labels and are used to emulate DFT relaxation before predicting a final energy. While the F1 scores and DAFs of energy-only models are surprisingly performant, their regression metrics such as $R^2$and RMSE are significantly worse. Of the energy-only models, only ALIGNN, BOWSR and CGCNN+P achieve positive coefficients of determination, R^2. Negative R^2 means model predictions explain the observed variation in the data less than simply predicting the test set mean. In other words, these models are not predictive in a global sense (across the full dataset range). However, even models with negative R^2 can be locally good in the positive and negative tails of the test set distribution. They suffer most in the mode of the distribution near the stability threshold of 0 eV/atom above the hull. This reveals an important shortcoming of R^2 as a metric for classification tasks like stability prediction.

The reason CGCNN+P achieves better regression metrics than CGCNN but is still worse as a classifier becomes apparent from fig. S5 by noting that the CGCNN+P histogram is more sharply peaked at the 0 hull distance stability threshold. This causes even small errors in the predicted convex hull distance to be large enough to invert a classification. Again, this is evidence to choose carefully which metrics to optimize. Regression metrics are far more prevalent when evaluating energy predictions. However, our benchmark treats energy predictions as merely means to an end to classify compound stability. Improvements in regression accuracy are of limited use to materials discovery in their own right unless they also improve classification accuracy. Our results demonstrate that this is not a given.

Figure 2 has models rank materials by model-predicted hull distance from most to least stable; materials fur-

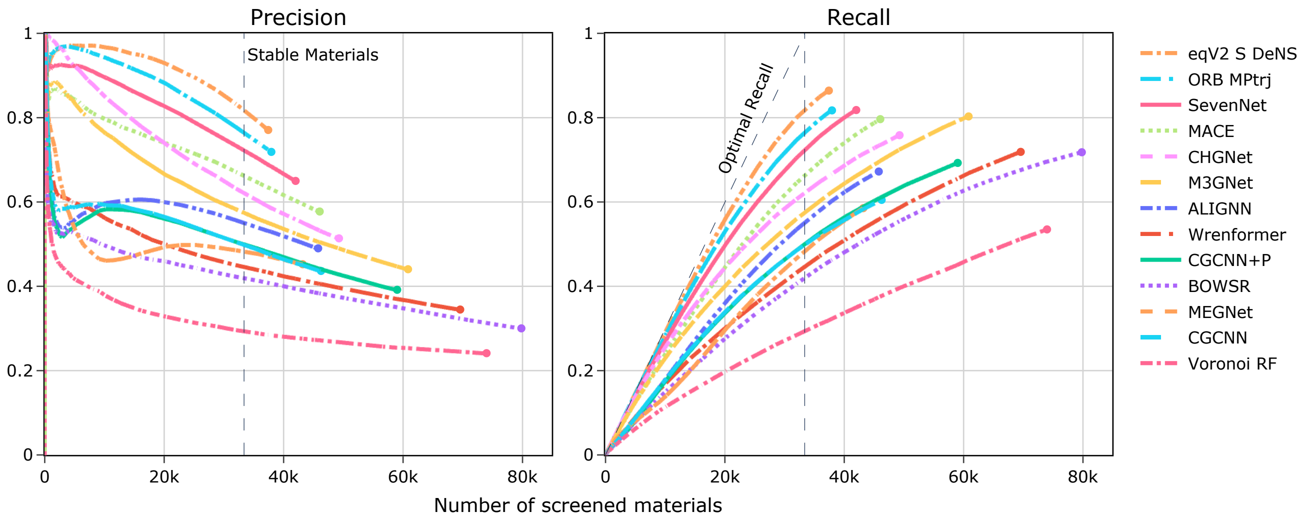

Figure 2: This figure measures model utility for materials discovery campaigns of varying sizes by plotting the precision and recall as a function of the number of model predictions validated. A typical discovery campaign will rank hypothetical materials by model-predicted hull distance from most to least stable and validate the most stable predictions first. A higher fraction of correct stable predictions corresponds to higher precision and fewer stable materials overlooked correspond to higher recall. Precision is calculated based only on the selected materials up to that point, whilst the cumulative recall depends on knowing the total number of positives upfront. This figure highlights how different models perform better or worse depending on the length of the discovery campaign. The UIPs are seen to offer significantly improved precision on shorter campaigns of ∼20k or less materials validated as they are less prone to false positive predictions among highly stable materials.

thest below the known hull at the top, materials right on the hull at the bottom. For each model, we iterate through that list and calculate at each step the precision and recall of correctly identified stable materials. This simulates exactly how these models would be used in a prospective materials discovery campaign and reveals how a model's performance changes as a function of the discovery campaign length. As a practitioner, you typically have a certain amount of resources available to validate model predictions. These curves allow you to read off the best model given these constraints. For instance, plotting the results in this manner shows that CHGNet initially achieves higher precision, than models such as EquiformerV2+DeNS, ORB MPTrj, SevenNet, and MACE that report higher precision across the whole test set.

In Figure 2 each line terminates when the model believes there are no more materials in the WBM test set below the MP convex hull. The dashed vertical line shows the actual number of stable structures in our test set. All models are biased towards stability to some degree as they all overestimate this number, most of all BOWSR by 133%. This is only a problem in practice for exhaustive discovery campaigns that validate all stable predictions from a model. More frequently, model predictions will be ranked most-to-least stable and validation stops after some pre-determined compute budget is spent, say, 10k DFT relaxations. In that case, the concentration of false positive predictions that naturally accumulates near the less stable end of the candidate list can be avoided with no harm to the campaign's overall discovery rate (see table S2 where even the DAF of the worst performing model benchmarked, Voronoi RF, jumps from 1.58 to 2.49).

The diagonal 'Optimal Recall' line on the recall plot in Figure 2 would be achieved if a model never made a false negative prediction and stopped predicting stable crystals exactly when the true number of stable materials is reached. Examining the UIP models, we find that they all achieve similar recall values, ranging from approximately 0.75 to 0.86.. This is substantially smaller than the variation we see in the precision for the same models ∼0.44-0.77. Inspecting the overlap, we find that the intersection of the models' correct agreements accounts for a precision of only 0.57 within the ∼0.75-0.86 range, with just 0.04 of the examples where all models are wrong simultaneously. These results indicate that the models are making meaningfully different predictions.

Examining the precision plot in Figure 2, we observe that the energy-only models exhibit a much more pronouced drop in their precision early-on, falling to 0.6 or less in the first 5k screened materials. Many of these models (all except BOWSR, Wrenformer and Voronoi RF) display an interesting hook shape in their cumulative precision, recovering again slightly in the middle of the simulated campaign between 5k and up to 30k before dropping again until the end.